

Headline

Estimated cost to run the full SkillsBench grid via the public Anthropic API on claude-sonnet-4-6 (with prompt caching): ~\$445 USD baseline; \$218-\$826 sensitivity range.

Main grid only (no baselines): \$390. Baselines only (noskill+gtonly): \$54.

Run scope (from run.sh --full + run_baselines.sh)

- Main grid: 84 SkillsBench tasks x 4 pool sizes (5/10/20/50) x 3 noise modes (random/hard/easy) x 1 seed = 1008 trials.
- Baselines: 84 tasks x 2 modes (noskill, gtonly), pool=0 = 168 trials.
- Grand total: 1176 agent invocations (minus tasks skipped because |GT skills| > pool_size, ~5%).

Empirical inputs (from already-completed runs)

Examined 3,110 trial records across 6 result files. After filtering out 1,473 1-second 'You've hit your limit' rate-limited entries (Max-plan throttling, not real runs), 1,182 real runs remain. Per pool size, the median observed values:

pool	n	sp_chars_med	wall_med_s	est_turns
0	265	373	258	21
5	214	12,937	250	21
10	278	14,210	236	20
20	209	23,218	248	21
50	216	42,613	264	22

Cost model

Pricing (claude-sonnet-4-6, public API, USD/M tokens):

input	\$3.00	output	\$15.00
cache write	\$3.75	cache read	\$0.30

Per-trial billing for one `claude -p` invocation with a growing cache prefix:

```

S = system_prompt_chars / 3.5 tokens
U = 600 tokens (task instruction)
T = 1200 tokens (avg Bash/Read tool result per turn)
O = 500 tokens (avg assistant output per turn)
N = wall_s / 12 (turn count, derived from wall time)

```

```
Turn 1: cache_write += S+U, output += O
```

```
Turn i>=2: cache_read += S+U+(i-2)*T, cache_write += T, output += O
```

```
cost = cache_write * $3.75/M + cache_read * $0.30/M + output * $15/M
```

Per-condition cost

pool	noise_modes	n_trials	\$/trial	\$/condition_group
5	3	252	\$0.3578	\$ 90.16
10	3	252	\$0.3411	\$ 85.96
20	3	252	\$0.3864	\$ 97.38
50	3	252	\$0.4635	\$ 116.80
0	2	168	\$0.3228	\$ 54.23

TOTAL		1176		\$ 444.53
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Aggregate token volume (baseline scenario)

cache writes :	35.64 M tokens	(\$ 133.66)
cache reads :	418.83 M tokens	(\$ 125.65)
output :	12.35 M tokens	(\$ 185.22)

total input :	454.48 M tokens
total output :	12.35 M tokens
TOTAL COST :	\$ 444.53

Sensitivity

scenario	\$total
Low (fewer turns, terse output)	\$ 218.42
Baseline (median observed)	\$ 444.53
High (long retries, verbose)	\$ 826.25

Low (fewer turns, terse output): Most tasks solved in <10 turns; agent emits compact summaries.

Baseline (median observed): Per-pool median wall-time; 12s/turn; 500 out-tok/turn.

High (long retries, verbose): Heavy debugging; agent verbose; cache miss on retries.

Caveats and how to anchor this number

- 1) Trial harness uses `claude -p --no-session-persistence --output-format text`, so no token counts were saved with the 967 real runs that already happened. Numbers above are model-based, not measured.
- 2) Real exact cost can be obtained cheaply by adding `--output-format json` to `call_claude()` in `testsets/exec_eval_prefill/run_trial.py`, then running ~10 representative trials. The JSON envelope includes `usage.input_tokens`, `output_tokens`, `cache_creation_input_tokens`, `cache_read_input_tokens` per trial. Multiply averages by the planned trial count for a measured estimate.
- 3) Pricing assumes claude-sonnet-4-6 public API rates with 5-min ephemeral prompt caching. Switching to 1-hour cache (6.00/M write) would raise cost ~15%; opting out of caching entirely would roughly 5-10x the input bill on long ReAct sessions.
- 4) Skipped trials (`|GT skills| > pool_size`) are not subtracted; that overstates cost by ~3-5%.
- 5) The friend's hint 'should be less than what you calculated' likely refers to: (a) prompt caching saves more than naive math implies on multi-turn ReAct, (b) median successful runs finish well before the `agent_wall_s` tail. Both factors are folded into the median-based baseline above; an even tighter estimate is in the Low scenario column.